

From Experimental Histories to Causal Surrogates: Data Reuse, Optimization, and Error Propagation

Research draft

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Abstract

A sequential experimental learner need not be the best model delivered after collection ends. We investigate whether retaining and refitting a completed experimental history improves the resulting causal surrogate, and which parts of the delivery procedure account for that change. A completed confirmation on one fixed deterministic emulator covers 12 acquisition histories: refitting all eligible paid observations reduces geometric mean snapped exact-level error to 0.183 times the online learner’s error (95% interval 0.102–0.327), with 11 histories improving. This comparison combines data reuse and additional optimization; it does not identify an architecture effect or an advantage at equal fitting compute. We give conditional statements for mechanism eligibility, deterministic error propagation and quantization sensitivity, with explicit counterexamples to monotonic improvement. Controlled attribution, independent-system comparisons and an archived physical case study remain necessary before submission. The present evidence supports a scoped post-collection delivery finding, without an acquisition or foundation-model claim.

1 Introduction

Experimental collection and final model delivery have different constraints. An online learner must update quickly and may discard observations to bound memory or computation. After collection, a separate fit can use retained observations, including paid probes and former validation rows. Any apparent benefit then has several possible explanations: a larger retained data set, more fitting compute, different optimization history, or the use of a known mechanism graph. Our aim is to separate these explanations.

The completed comparison supplies evidence for a delivery recipe on one emulator. The experimental program below is designed to test whether that recipe has a distinct causal-surrogate contribution. An ordinary flat refit explaining the result would narrow the contribution to delivery and experimental accounting. We will retain that conclusion rather than infer a more general architectural advantage from a favorable aggregate.

Current contributions and scope. We report a completed, receipt-backed comparison across repeated acquisition histories of one fixed system; derive conditions and limitations relevant to reuse and chain prediction; and implement an attribution protocol that separates retained data from fitting budget. Independent-system and physical results are pending. Neither causal structure discovery nor learned experiment selection is a current contribution. No foundation model is used in the delivery experiment.

2 Problem and related work

We assume a known directed acyclic graph. A surrogate predicts each nonroot node from its measured parents during fitting and its predicted parents during free-running deployment. Correct interventions replace the relevant prediction with its clamped value. A roots-to-target flat predictor supplies a simpler prediction control; when intermediate mechanisms are intervened on, a flat predictor without action inputs cannot represent that response using roots alone. Such rows are excluded from its observational-map fit.

The connection between neural models and SCMs already has substantial theory. Expressiveness is distinct from causal learnability, and structural constraints and identification assumptions matter (Xia et al., 2021). We do not claim that neural approximation or low MSE identifies an unrestricted SCM. Our conditional mechanism regression and deterministic propagation statements address a much narrower known-graph setting. Causal Chambers provide an external physical apparatus with published causal models (Gamella et al., 2025); the planned archive study assesses physical prediction under that apparatus’s conditions, without claiming new live experiment design.

3 Completed delivery confirmation

The comparison retains the originally registered 12 histories of a single deterministic emulator. The delivery procedure fits separate mechanism MLPs from all eligible paid rows using full-batch Adam and additional epochs. Original online weights remain unchanged. Evaluation runs the learned chain on the full previously exposed discrete grid using predicted intermediate values. Acquired input points occur within this grid, so it is not a held-out prospective sample.

The primary endpoint is exact-level error after mapping predictions to the nearest target level. For each history, the registered numerator is the median of three scored optimization initializations, and the denominator is the original online chain’s error. That median summarizes sensitivity; it is not a deployable rule for selecting a model without access to true test outcomes. Histories, not initialization repeats or grid rows, are the paired units.

The geometric mean paired ratio is 0.183, with a log-scale t interval of $[0.102, 0.327]$ and exhaustive two-sided sign-flip $p = 0.00146484$. 11 of 12 histories improve. The worsening history 124753321 changes from 0.196339 online error to 0.395658 delivery median error (ratio 2.015177); it is retained without exclusion or rescue tuning. The generated values and source hashes are listed in `claim_index.json`.

This supports the bundled delivery procedure on repeated histories of that emulator. Paid histories are matched, but fitting compute is not. It establishes neither acquisition superiority, causal-factorization attribution, a foundation model benefit, nor generalization over independently parameterized systems.

4 Attribution and prospective evaluation protocol

Stage A: attribution (exploratory; pending). We compare three unique-row data sets: the final 50-entry buffer, all rows actually admitted at live online updates, and all eligible paid rows. Each is fit from scratch for 100 or 30,000 epochs with fixed architecture, initialization scheme, optimizer and intervention masks. The original online model is a separate comparator. Flat MLPs have both equal-epoch and separately labeled matched-CPU comparisons. Fixed ridge, polynomial and tree controls are configured on an earlier development history, with no configuration tuning on the twelve exposed histories.

Replay reconstruction reveals additional differences from the online training path: the selected observation is duplicated in consolidation; per-node fast adaptation can include a directly intervened node; and the final observation refresh enters the saved buffer after the last fit. These differences are recorded rather than silently attributed to data quantity. The scratch factorial controls masks among its cells but does not exactly replay the online objective. Diagnostics retain support coverage, observed-parent residuals, propagated prediction shifts and decision margins, especially for the worsening history. No simulator responses are acquired in Stage A.

Stage B: controlled generalization (gated; pending). After attribution, freeze a deployable recipe with initialization zero, the strongest simpler refit and a decisive ablation. Compare them and the online learner on independently parameterized five-node and thirty-node systems in the audited generator families. Both balanced varied-value and matched random histories are shared across prediction methods. Complete joint-action blocks are reserved before collection. Continuous held-out normalized MSE is primary; snapped error is secondary only for genuinely quantized outputs. Histories and optimization repeats are nested within systems. Paired log-ratio contrasts, multiple-testing correction, practical thresholds and separate graph-size reporting are frozen before opening outcomes. The stochastic estimand must be specified explicitly: composing conditional means through nonlinear downstream functions does not generally produce their interventional expected outcome.

Stage C: archived physical prediction (pending). Use the eleven previously unscored conditions of `lt_malus_v1`; `white_64` remains development only. Duplicate commands and joint action blocks are grouped before splitting. Compare delivery, rolling-buffer learning, joint Fourier regression and the published relative-angle physics model, using continuous prediction error. The physics basis is an intercept and $\cos^2(\theta_1 - \theta_2)$, with degree inputs converted to radians; the earlier product-form attenuation control is not retained. Conditions and repeated readings from one apparatus are not independent physical worlds. All conditions and conditional uncertainty will be reported, including negative comparisons against physics.

5 Conditions and limits of delivery

We study a known acyclic graph with measured parents. The following statements explain permissible reuse and possible error amplification. They do not prove identification for arbitrary neural models or certify the fitted networks.

Proposition 5.1 (Mechanism eligibility). *Let $V_i = f_i(V_{\text{Pa}(i)}) + U_i$ be an invariant additive-noise mechanism. Suppose U_i is integrable, independent of the other exogenous variables, and has mean zero. An intervention policy may depend on previous experiments, but must choose its action before the current exogenous noises are realized. Retain a row only when the action does not replace mechanism i , and when retention does not select on its current noise or response conditional on its parents. Then, in the retained interventional mixture,*

$$\mathbb{E}[V_i \mid V_{\text{Pa}(i)} = p, \text{retained}] = f_i(p).$$

Thus squared-loss regression can target this mechanism on supported parent values, assuming finite second moments. This is an estimand statement; finite data, optimization and extrapolation guarantees require further assumptions.

Proof. In an acyclic graph, a node's parents depend only on earlier exogenous noises and the chosen action. The current U_i is independent of those noises and of past experiments. Since the policy precedes the current realization, and the retention rule does not condition on U_i , the conditional expectation of U_i remains zero. Substituting the structural equation yields the assertion. The conditional expectation minimizes population squared loss on the retained support. No statement outside that support follows. $\square$

Confounding, missing parents, interventions that change rather than clamp a mechanism, and response-dependent retention can invalidate the premise. In particular, selecting the best probe using its measured response can change the noise distribution. All-paid reuse avoids discarding probes on that basis, but still requires invariant mechanisms and measured parents. The current confirmation uses a deterministic emulator; this distinction matters for the prospective noisy systems. A non-additive stochastic SCM also needs a distributional treatment: composing conditional means need not produce a downstream interventional mean through a nonlinear mechanism.

Proposition 5.2 (Deterministic DAG propagation). *Consider a deterministic SCM and its learned functions $\hat{f}_i$, evaluated in topological order under identical correctly clamped interventions. For each non-clamped node, assume $|\hat{f}_i(v_{\text{Pa}(i)}) - f_i(v_{\text{Pa}(i)})| \leq \epsilon_i$ at every true parent vector in the deployment support. Assume also*

$$|\hat{f}_i(p) - \hat{f}_i(q)| \leq \sum_{j \in \text{Pa}(i)} L_{ij} |p_j - q_j|$$

on a region containing both true and predicted parent vectors, with $L_{ij} \geq 0$. Writing $e_i = |\hat{v}_i - v_i|$, one has

$$e_i \leq \epsilon_i + \sum_{j \in \text{Pa}(i)} L_{ij} e_j.$$

Correctly clamped nodes have $e_i = 0$. If their rows in L and their local bounds are set to zero, the vector bound is $e \leq (I - L)^{-1} \epsilon = \sum_{k=0}^{d-1} L^k \epsilon$, where d is the number of nodes and L is strictly triangular in topological order.

Proof. Add and subtract $\hat{f}_i(v_{\text{Pa}(i)})$ and apply the triangle inequality. The local approximation bounds one term; the coordinate Lipschitz inequality bounds the other. Clamps are identical by assumption. Substitution in topological order gives the path sum. A strictly triangular $d \times d$ matrix satisfies $L^d = 0$, so the finite geometric sum is its inverse. $\square$

The bound can amplify a small local error along deep or high-gain paths. Observed-parent test MSE is neither a uniform ϵ_i nor a verified Lipschitz constant. Our residual and support diagnostics are explanatory measurements, not certified instances of this proposition.

Proposition 5.3 (Quantization margin). *For distinct ordered levels $\ell_1 < \dots < \ell_m$, let Q assign the nearest level, with a fixed tie rule, and let $b_k = (\ell_k + \ell_{k+1})/2$. Define $M(y) = \min_k |y - b_k|$. If $|\hat{y} - y| < M(y)$ then $Q(\hat{y}) = Q(y)$. If a pointwise prediction bound $B(y)$ is valid, $\Pr(Q(\hat{y}) \neq Q(y)) \leq \Pr(M(y) \leq B(y))$.*

Proof. The open interval of radius $M(y)$ around y crosses no decision boundary. Consequently both values occupy the same decision cell. The contrapositive implies that a quantization error requires $M(y) \leq |\hat{y} - y| \leq B(y)$. Strict inequality excludes boundary ties. $\square$

Snapped error depends on the location of errors relative to decision boundaries; similar MSE alone need not imply similar exact-level accuracy.

Counterexamples. For a constant predictor and test response zero, fitting a single training response zero gives zero risk; adding a noisy response two makes its least-squares mean one and test risk one. With a noisy training target one, optimizing a constant from 0.1 to one reduces training error but increases test error. For a two-stage chain, suppose the true parent and child are zero on the observed support, the learned parent predicts 0.1, and the learned child is $\hat{f}(z) = 100z$. The child has zero observed-parent error at zero, but its free-running prediction is ten. These are existence examples, not descriptions of the measured histories. Analytically solved chains, forks, colliders and clamps are checked by executable tests.

6 Reproducibility and resource accounting

Fitting and evaluation run in separate processes. Training inputs, row selections, source, dependency versions, optimizer configuration and outputs are hash-bound. Fit workers are denied network access and evaluation arrays. Exclusive output paths preserve interrupted attempts and reject replacement. Supervisors enforce wall-time and memory ceilings. CPU time, wall time, unique eligible rows, parameters, optimizer updates and memory accompany each fit. Original histories and weights are retained under local result custody. The claim generator binds every retained numerical result to its validated receipt. Experimental controls, analytic checks and the execution ledger are versioned with the manuscript. No closed-source API is needed for the critical path.

7 Limitations and conclusion

The current positive result is restricted to a bundled delivery recipe, a quantized endpoint, and repeated histories of a single fixed emulator on an exposed evaluation grid. It does not yet justify a new architecture or equal- compute superiority. The theoretical statements require graph, invariance, noise and support assumptions that cannot be inferred from small fitted residuals. Additional observations and optimization can worsen test performance, and local accuracy can fail under chain deployment.

Submission requires an attributable benefit, prospective controls, an honest external boundary, compute accounting and complete reproducibility. If ordinary refitting or matched compute removes the distinction, a narrower empirical methodology report is the appropriate outcome. The present draft makes those unresolved requirements explicit.

References

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